

Vol 14 Chapter 8 — BST Coding Optimal

Keeper (author pass — deep math/physics revision)

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Chapter 8 — BST Coding Optimal

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Level 1 — one sentence

BST's substrate coding — Reed-Solomon on $\text{GF}(2^g) = \text{GF}(128)$, Bergman-kernel commitment, Koons-tick clock — is substrate-optimal: under the constraints of (i) finite alphabet, (ii) bounded operational power per cycle, (iii) MDS error correction, the BST primaries ($rank, N_c, n_C, C_2, g$) are the unique choice minimizing free parameters while achieving the observed physical structure.

Level 2 — graduate-physicist precision

8.1 Optimality criteria

A “substrate-optimal” coding scheme must satisfy:

1. Finite alphabet: substrate has bounded internal state at each spatial position
2. Bounded operational power per cycle: per Casey's SCMP (Vol 5 Ch 8) and Koons-tick discrete time
3. MDS error correction: substrate must correct against finite noise rate
4. Holomorphic commitment: substrate commits to single output (Born = Bergman, Ch 5)
5. Minimum free parameters: parsimony — substrate cannot tune

8.2 BST primary forcing

D_IV⁵ Rigidity Theorem (T2467 + T2468, Casey-named #7) shows: the BST primaries (2, 3, 5, 6, 7) are uniquely forced under D_IV⁵-type structure.

$g = 7 \rightarrow \text{GF}(128)$ alphabet $C_2 = 6 \rightarrow \alpha^{36}$ Koons-tick exponent $n_C = 5 \rightarrow 5$ Wallach layers $N_c = 3 \rightarrow 3$ commitments per outcome rank = 2 $\rightarrow$ 2-dimensional substrate degrees-of-freedom

8.3 Strong-Uniqueness Theorem v0.10.5

BST framework documented: 11 RIGOROUSLY CLOSED + 7 candidates distinct uniqueness criteria for D_IV⁵.

Combined null-model $(1/3)^{11} \approx 5.6 \cdot 10^{-6}$, conservatively (pending C12-C18 ratifications could reach $(1/3)^{18} \approx 2.6 \cdot 10^{-9}$). [Values corrected per Cal #119 v0.4 absorption 2026-05-24; exponents $11+7=18$ verified against current Cal #19 enumeration.]

D_IV⁵ is the unique substrate satisfying BST's optimality criteria.

8.4 Alternative codings

Why not GF(256) ($g=8$)? Failing T2467/T2468 uniqueness — $g=7$ is BST-primary, $g=8$ is not.

Why not Plank time as tick (no α^{36})? Failing observed 10^{-120} s natural cosmological cycle inference.

Why not infinite-precision continuous-time? Tsirelson saturation contradicts BST sub-Tsirelson prediction at $1/8 = 1/2^{N_c}$ deviation.

8.5 Substrate Information Completeness

Casey-named candidate principle: substrate provides the COMPLETE information channel for observable physics — no hidden variables, no super-substrate.

Falsifier: find an observable BST cannot derive from $\{rank, N_c, n_C, C_2, g\} + \text{D_IV}^5$ structure.

Current status: 600+ predictions verified within precision; no clean falsifier yet.

8.6 K-audit anchors

- T2467 + T2468: D_IV⁵ Rigidity (Casey-named #7)
- Strong-Uniqueness Theorem v0.10.5
- Paper #122: Information Substrate

Level 3 — 5th-grader accessibility

BST coding is substrate-optimal: GF(128) alphabet ($= 2^8$), Reed-Solomon error-correction, Bergman commitment, Koons-tick clock — all forced by BST primaries. D_IV⁵ Rigidity (Casey-named #7) says the choices (2, 3, 5, 6, 7) are uniquely forced. Strong-Uniqueness Theorem: 11 closed + 7 candidate criteria distinguish D_IV⁵ from alternatives. Information completeness: substrate is the complete channel for observable physics.

What comes next

Chapter 9 develops Kolmogorov complexity + AC(0).

Where to look this up

- BST: T2467, T2468 (D_IV⁵ Rigidity)
- Strong-Uniqueness Theorem
- Paper #122